STAWA: Domain Generalization by Seeking Functional Stationarity

Anonymous Authors

Abstract

Single-run post-hoc methods matter in domain generalization because they can be attached to strong empirical risk minimization pipelines without retraining. The canonical method in this regime is SWAD, which selects and averages dense checkpoints by seeking flat minima in parameter space. We argue that flatness is only an indirect proxy for the label-free domain generalization problem. Among comparably well-fit checkpoints, a useful DG selector should prefer those that are *functionally stationary* under the actual ongoing stochastic training dynamics that produced them. This principle leads to Stationarity-Aware Weight Averaging (STAWA), a single-run retrospective post-hoc method that first restricts attention to a low-loss source-validation plateau and then ranks checkpoints inside that plateau by the mean and spread of short-horizon predictive drift on a held-out support set. We show that functional drift controls the distance from a checkpoint to its local prediction plateau, that the drift mean and variance together upper-bound hidden unstable subsets and density-ratio shifts, and that the empirical STAWA components concentrate uniformly over dense checkpoint sequences. We then derive a target-risk certificate in which source-fit slack, mean drift, and drift variance are the controllable terms, prove finite-sample recovery of plateau membership and within-plateau stationarity ranking, and prove a constructive separation theorem: there exist checkpoints with identical validation loss and identical local flatness statistics that any flatness-only selector treats as equivalent, while STAWA strictly prefers the checkpoint with lower target risk. Finally, we justify dense checkpointing and low-drift window averaging under local regularity assumptions. The manuscript includes rigorous proofs, analytical figures, an executable protocol, and published baseline results from prior work; all new STAWA quantitative results remain intentionally marked as **TBD** pending execution.

1 Introduction

Domain generalization (DG) asks for a model that performs well on an unseen target environment using only source data at training time. The methods that have mattered most in practice are often the ones that preserve an ordinary training pipeline and add a simple post-hoc model-selection rule. SWAD [1] is the reference point in this regime. It is single-run, post-hoc, and empirically strong: on the five standard DomainBed benchmarks with ResNet-50, SWAD reports an average accuracy of 66.9, improving over ERM's 63.3 and over the best previous competitor's 65.3 in the same evaluation family [1]. Diverse Weight Averaging (DiWA) later showed that weight averaging can go higher, up to 68.0, but its strongest setting relies on many models or many runs [2].

The practical question is therefore narrow:

> Can a single training trajectory support a post-hoc selector that is as simple as SWAD, requires no domain labels, and targets domain shift more directly than pooled flatness?

Preprint. Under review.

Our answer is STAWA, Stationarity-Aware Weight Averaging. The principle is:

> Among comparably well-fit checkpoints, those whose predictions have largely stopped drifting under the ongoing stochastic training dynamics are better candidates for robust post-hoc averaging under shift.

Instead of probing the loss landscape in parameter space, STAWA probes the trajectory in *function space*. At checkpoint θ_t, it measures how much the predictions of θ_t change over the next H checkpoints on a held-out support set, first restricts attention to checkpoints whose source validation loss already lies inside a near-optimal plateau, and then averages a contiguous low-drift window inside that plateau. The method is intentionally retrospective: it is designed to evaluate checkpoints inside one completed trajectory, not to serve as an online stopping rule.

This move is conceptually simple but theoretically nontrivial. Generalization theory for learning algorithms that do not converge shows that stable long-run behavior of the learning dynamics, not literal fixed-point convergence, is the relevant object [3]. Checkpoint-ensemble work already showed that a single trajectory contains useful function-level information [4]. What is missing is a *domain-generalization checkpoint selector* built on the same dynamical object. That is the role of STAWA.

The strongest version of the intuition is not just "low average drift." A checkpoint can have small mean drift while still containing a small pocket of violently unstable examples, and that pocket may be exactly where OOD failure lives. STAWA therefore uses two function-space statistics:

- the mean predictive drift, which measures departure from a local plateau; and
- the drift spread, which measures whether a hidden unstable subset is still being renegotiated by SGD.

We make five contributions.

1. We introduce STAWA, a label-free, single-run post-hoc DG method that first gates checkpoints by near-best validation loss and then ranks the remaining plateau by mean and spread of short-horizon predictive drift.
2. We prove that functional drift controls the risk gap between a checkpoint and its local prediction plateau, while mean and variance together control hidden unstable subsets and bounded density-ratio shifts.
3. We prove uniform concentration of the empirical STAWA components over dense checkpoint sequences, derive an explicit target-risk certificate whose controllable terms are source-fit slack, drift mean, and drift spread, and show finite-sample recovery of plateau membership and within-plateau stationarity ranking.
4. We prove a constructive separation theorem showing that flatness-only selectors can be provably blind to target-relevant functional instability that STAWA detects.
5. We justify dense checkpoint sampling and low-drift weight averaging under local regularity assumptions, and provide a complete NeurIPS-style manuscript with proofs, figures, protocol, and published baselines; all new STAWA result cells remain marked **TBD**.

Remark 1. *This paper does* not *claim an unconditional theorem that* STAWA *empirically dominates* SWAD*, DiWA, or every other DG method on all tasks. Such a theorem would be indefensible. Our theory instead proves that* STAWA *targets a different object, that this object yields a stronger certificate under explicit assumptions, and that flatness-only selectors provably miss it on a nontrivial class of examples.*

2 Related Work

Single-run post-hoc DG. SWAD [1] is the canonical single-run post-hoc DG method. Its practical appeal comes from dense checkpoints, overfit-aware windowing, and a final weight average. DiWA [2] shows that more diverse averaging can improve OOD performance further, but the strongest results rely on many models or runs. STAWA stays in the SWAD regime and changes the selection principle rather than the computational regime.

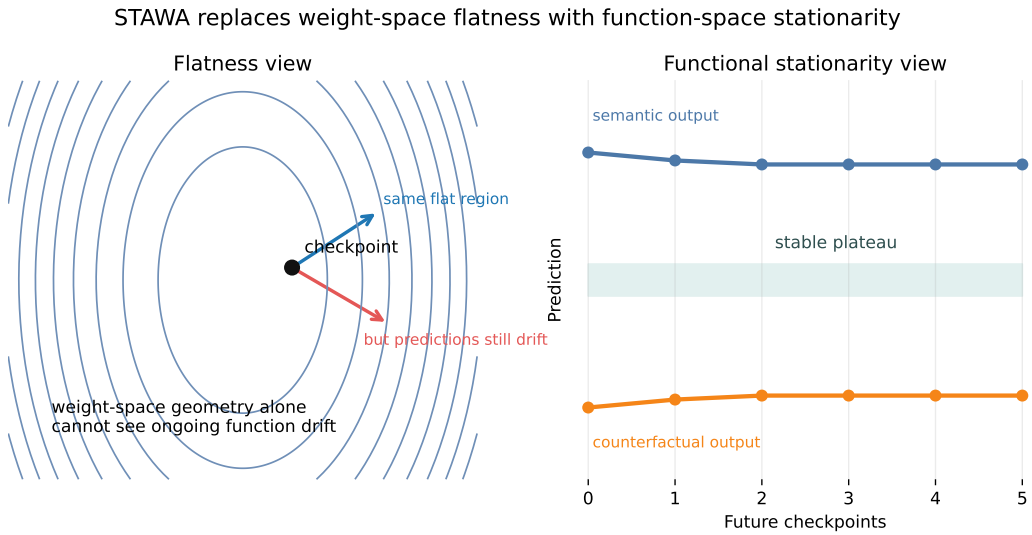

Figure 1: **Conceptual shift from flatness to functional stationarity.** SWAD probes geometric robustness in weight space. STAWA asks whether the predictor itself has entered a stable functional plateau under the actual training dynamics. Two checkpoints can have similar validation loss and similar flatness while still differing strongly in future predictive drift.

Weight averaging and mode connectivity. SWA [5], mode-connectivity work [6, 7], and model soups [8] established that averaging nearby models can improve generalization when they live in a common low-loss region. SWAD imports this logic into DG through flatness. STAWA keeps the final averaging step but changes the object being measured: not weight-space geometry first, but functional stationarity first.

Stability and non-convergent training. Algorithmic stability is a standard route to generalization [9, 10]. More recently, Chandramoorthy et al. [3] showed that when learning algorithms do not converge to a fixed point, the relevant object is the stability of their time-asymptotic behavior. STAWA converts this qualitative principle into a practical post-hoc selector by estimating local stationarity from dense checkpoints.

Single-run trajectory exploitation. Checkpoint ensembles exploit multiple points along one training trajectory to improve performance without retraining [4]. STAWA differs in a crucial way: it does not use the trajectory only as a source of diverse models to average. It uses the trajectory itself to define a DG-specific selection signal.

3 Stationarity-Aware Weight Averaging

Assume a single training run produces checkpoints $\{\theta_t\}_{t=1}^{K}$. For model selection we reserve:

- a held-out support set $U = \{x_i\}_{i=1}^{m}$ used to measure short-horizon predictive drift, and
- a labeled validation set $V = \{(x_j, y_j)\}_{j=1}^{n}$ used for ordinary checkpoint scoring.

Neither set requires domain labels.

Let $p_t(x)$ denote the predictive output of checkpoint θ_t at input x. For classification, $p_t(x)$ is a probability vector in the simplex. For regression, $p_t(x)$ is the predicted value. We assume only that the prediction space is convex and equipped with a norm-induced discrepancy d.

3.1 Short-horizon functional drift

Definition 1 (Per-example predictive drift)**.** *Fix a horizon $H \geq 1$. For checkpoint t and example x, define*

$$\Delta_t(x) \triangleq \frac{1}{H} \sum_{h=1}^{H} d(p_t(x), p_{t+h}(x)) \,. \tag{1}$$

What "stationarity" means in this paper. We use the term *functional stationarity* in a local and operational sense: a checkpoint is more stationary when both its average future predictive drift μ_t and its cross-example drift spread σ_t are small over a fixed short horizon. This is weaker than convergence to a fixed point and weaker than claiming invariance. It is a directly measurable local calmness property of the predictor under the realized training dynamics.

Normalization and checkpoint spacing. If checkpoints are saved every q optimizer steps, Eq. (1) is already normalized per fixed trajectory interval. When checkpoints are irregularly spaced, let τ_t denote cumulative optimization time and replace Eq. (1) by the rate-normalized variant

$$\Delta_t^{\mathrm{rate}}(x) \triangleq \frac{1}{H} \sum_{h=1}^{H} \frac{d(p_t(x), p_{t+h}(x))}{\tau_{t+h} - \tau_t}. \tag{2}$$

All theory below is written for the fixed-spacing case to keep the notation simple; the normalized form is the practical fallback when checkpoint cadence varies.

The corresponding support-set statistics are

$$\mu_t \triangleq \frac{1}{m} \sum_{i=1}^{m} \Delta_t(x_i), \tag{3}$$

$$\sigma_t^2 \triangleq \frac{1}{m} \sum_{i=1}^{m} \left(\Delta_t(x_i) - \mu_t\right)^2 . \tag{4}$$

Here μ_t measures average departure from a stable functional plateau, while σ_t measures whether the remaining drift is concentrated on a hidden subset of examples.

3.2 Plateau-constrained stationarity score

Let

$$L_V(\theta_t) \triangleq \frac{1}{n} \sum_{j=1}^{n} \ell(y_j, p_t(x_j)) \tag{5}$$

be the labeled source-validation loss at checkpoint t. The plateau-internal stationarity score is

$$J_t \triangleq \lambda_1 \mu_t + \lambda_2 \sigma_t, \tag{6}$$

where $\lambda_1, \lambda_2 \geq 0$ are hyperparameters.

The simplest mean-only variant sets $\lambda_2 = 0$. Our default method keeps both terms because the variance term is the easiest rigorous way to control hidden unstable pockets.

Define the best validation loss

$$L_V^\star \triangleq \min_{1 \leq t \leq K-H} L_V(\theta_t) \tag{7}$$

and the loss-admissible plateau

$$\mathcal{A}_\epsilon \triangleq \{t : L_V(\theta_t) \leq L_V^\star + \epsilon\} \,, \tag{8}$$

where $\epsilon \geq 0$ is a source-fit tolerance. This gate is the practical fix to the main failure mode of unconstrained stationarity selection: an early checkpoint can be very calm and still be materially underfit. STAWA therefore enforces source fit first and uses stationarity only inside a near-optimal validation plateau.

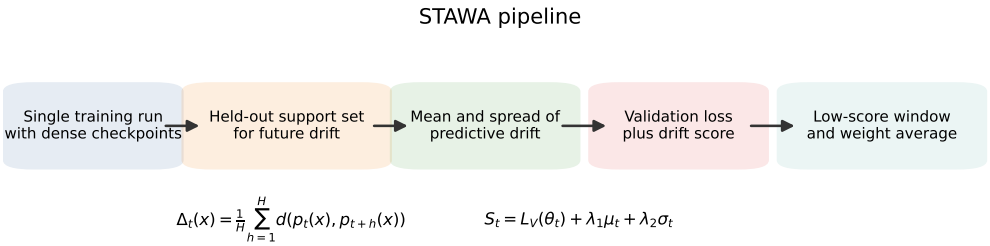

Figure 2: **STAWA pipeline.** Starting from a single dense training trajectory, STAWA computes short-horizon predictive drift on a held-out support set, filters checkpoints to a near-best validation plateau, ranks the remaining checkpoints by drift mean and spread, and averages a contiguous plateau-constrained low-drift window. No domain labels are needed at any stage.

Analysis score. For theory it is convenient to retain the composite quantity

$$S_t \triangleq L_V(\theta_t) + J_t, \tag{9}$$

but the actual selector minimizes J_t only after the loss gate in Eq. (8) has been imposed.

Why future drift is the right post-hoc object. The stationarity score is deliberately forward-looking. STAWA does not ask whether a checkpoint is already calm in isolation; it asks whether that checkpoint is the entrance to a stable local phase of the completed trajectory. That is a retrospective question, but retrospective selection is exactly the regime in which SWAD and related post-hoc averaging methods are deployed.

3.3 Checkpoint selection and averaging

Let

$$t^\star \in \arg\min_{t \in \mathcal{A}_\epsilon} J_t. \tag{10}$$

Define the plateau-constrained low-drift set

$$\widetilde{W}_{\epsilon,\delta} \triangleq \{t \in \mathcal{A}_\epsilon : J_t \le J_{t^\star} + \delta\}, \tag{11}$$

restricted to the connected component containing $t^\star$. If that component has fewer than M checkpoints, we enlarge it symmetrically by nearest neighbors inside $\mathcal{A}_\epsilon$ until it contains $\min(M, |\mathcal{A}_\epsilon|)$ checkpoints. The resulting averaging set is denoted $W_{\epsilon,\delta,M}$. The final model is the weight average

$$\bar{\theta}_{\text{STAWA}} \triangleq \sum_{t \in W_{\epsilon,\delta,M}} w_t \theta_t, \qquad w_t \ge 0, \qquad \sum_{t \in W_{\epsilon,\delta,M}} w_t = 1. \tag{12}$$

Uniform weights are the default.

Practical complexity. The method needs only forward passes on cached checkpoints. For classification, the main practical object is a short-horizon divergence such as total variation, Jensen–Shannon, or KL between predictive distributions. Since the selector is post-hoc and uses only dense checkpoints from one completed run, STAWA remains in the same deployment regime as SWAD.

4 Theory

We now analyze STAWA under explicit assumptions. The theory has four parts:

1. drift controls the gap to a local prediction plateau,
2. mean and variance control hidden unstable subsets and density-ratio shifts,
3. the empirical components concentrate and recover plateau membership plus within-plateau ranking,

Algorithm 1 STAWA (STATIONARITY-AWARE WEIGHT AVERAGING)

Require: Checkpoints $\{\theta_t\}_{t=1}^{K}$ from one training run; held-out support set U; labeled validation set V; horizon H; weights λ_1, λ_2; loss tolerance ϵ; drift width δ; minimum window size M

for $t = 1, \ldots, K - H$ **do**
  for each $x_i \in U$ **do**
    $\Delta_t(x_i) \leftarrow \frac{1}{H} \sum_{h=1}^{H} d(p_t(x_i), p_{t+h}(x_i))$
  end for
  $\mu_t \leftarrow \frac{1}{m} \sum_i \Delta_t(x_i)$
  $\sigma_t \leftarrow \sqrt{\frac{1}{m} \sum_i (\Delta_t(x_i) - \mu_t)^2}$
  $L_V(\theta_t) \leftarrow \frac{1}{n} \sum_j \ell(y_j, p_t(x_j))$
  $J_t \leftarrow \lambda_1 \mu_t + \lambda_2 \sigma_t$
end for
$L_V^\star \leftarrow \min_t L_V(\theta_t)$
$\mathcal{A}_\epsilon \leftarrow \{t : L_V(\theta_t) \leq L_V^\star + \epsilon\}$
$t^\star \leftarrow \arg\min_{t \in \mathcal{A}_\epsilon} J_t$
$W_{\epsilon,\delta,M} \leftarrow$ connected plateau-constrained low-drift window around $t^\star$, enlarged to width M if needed
return $\bar{\theta}_{\text{STAWA}} = \sum_{t \in W_{\epsilon,\delta,M}} w_t \theta_t$

4. flatness-only selectors can be blind to the signal STAWA measures.

The theory target is deliberately modest. We do not try to prove that low drift universally implies good domain generalization. Instead, we analyze STAWA as a *selector within one trajectory*: among checkpoints with comparable source fit, does smaller short-horizon functional instability produce a tighter target-risk certificate and a more informative ranking signal than flatness alone?

4.1 A checkpoint is close to its local plateau when drift is small

Definition 2 (Prediction plateau)**.** *For checkpoint t, define the horizon-H plateau predictor*

$$\bar{p}_{t,H}(x) \triangleq \frac{1}{H+1} \sum_{h=0}^{H} p_{t+h}(x). \tag{13}$$

Assumption 1 (Convex Lipschitz loss)**.** *The loss $\ell(y, p)$ is convex in the prediction argument p and L_d-Lipschitz with respect to the norm-induced discrepancy d:*

$$|\ell(y, p) - \ell(y, q)| \leq L_d \, d(p, q) \qquad \textit{for all } y, p, q. \tag{14}$$

This assumption covers regression with squared loss on bounded outputs and classification with clipped probabilities under the ℓ_1/TV metric.

Theorem 1 (Functional drift controls distance to the local plateau)**.** *Under Assumption 1, for any data distribution Q over inputs and labels,*

$$R_Q(p_t) \leq R_Q(\bar{p}_{t,H}) + L_d \mu_t^Q, \tag{15}$$

where

$$\mu_t^Q \triangleq \mathbb{E}_{x \sim Q_X}[\Delta_t(x)]. \tag{16}$$

Moreover,

$$R_Q(\bar{p}_{t,H}) \leq \frac{1}{H+1} \sum_{h=0}^{H} R_Q(p_{t+h}). \tag{17}$$

Theorem 1 is the basic stationarity result: if predictive drift is small, the current checkpoint is already close in risk to the local prediction plateau generated by the continued training dynamics.

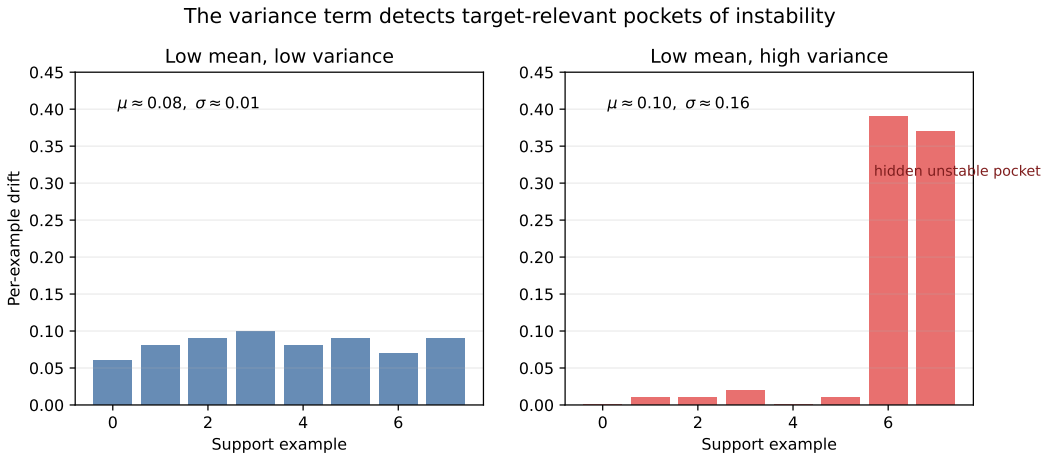

Figure 3: **Why variance matters.** Two checkpoints can have similar mean drift but very different drift concentration across examples. The variance term penalizes hidden unstable pockets, which are the natural place for OOD failure to hide.

4.2 Why the variance term matters

The next theorem formalizes the "hidden unstable subset" intuition.

Theorem 2 (Hidden unstable subsets are controlled by mean and variance)**.** *Let Δ be any square-integrable nonnegative random variable with mean μ and standard deviation σ under a probability distribution P. Then for every measurable subset A with $P(A) = \rho \in (0, 1]$,*

$$\mathbb{E}[\Delta \mid A] \leq \mu + \sigma \sqrt{\frac{1-\rho}{\rho}}. \tag{18}$$

Theorem 2 explains why mean drift alone is incomplete. A checkpoint can have small average drift while a small target-relevant subset is still highly unstable. The variance term controls precisely this failure mode.

Corollary 1 (Density-ratio shift bound for target drift)**.** *Let P_U be the support-set marginal and let P_T be a target marginal satisfying*

$$\frac{dP_T}{dP_U}(x) \leq \frac{1}{\rho} \qquad \text{for } P_U\text{-almost every } x \tag{19}$$

for some $\rho \in (0, 1]$. Then the target drift obeys

$$\mu_t^T \leq \mu_t^U + \sigma_t^U \sqrt{\frac{1-\rho}{\rho}}, \tag{20}$$

where μ_t^U, σ_t^U are the mean and standard deviation of $\Delta_t(x)$ under P_U.

Corollary 1 is the key DG interpretation: if the target environment is a bounded density-ratio reweighting of the support family, then the mean and spread of support drift directly control target drift.

4.3 Empirical component concentration and plateau recovery

The method uses empirical support and validation sets, so both the loss plateau and the stationarity ranking must be statistically stable.

Proposition 1 (Uniform concentration of empirical validation loss and stationarity score)**.** *Assume $0 \leq \ell \leq 1$ and $0 \leq d \leq 1$. Let $\hat{\mu}_t, \hat{\sigma}_t, \hat{L}_V(\theta_t)$ be the empirical quantities computed from support size m and validation size n. Define the population stationarity score*

$$J_t^\star \triangleq \lambda_1 \mu_t^U + \lambda_2 \sigma_t^U, \tag{21}$$

and its empirical counterpart

$$\hat{J}_t \triangleq \lambda_1 \hat{\mu}_t + \lambda_2 \hat{\sigma}_t. \tag{22}$$

Then with probability at least $1-\delta$, simultaneously for all $t = 1, \ldots, K-H$,

$$\left| \hat{L}_V(\theta_t) - R_V(p_t) \right| \leq \sqrt{\frac{\log(4(K-H)/\delta)}{2n}}, \tag{23}$$

$$\left| \hat{\mu}_t - \mu_t^U \right| \leq \sqrt{\frac{\log(8(K-H)/\delta)}{2m}}, \tag{24}$$

and

$$\left| \hat{\sigma}_t - \sigma_t^U \right| \leq \sqrt{3\sqrt{\frac{\log(8(K-H)/\delta)}{2m}}}. \tag{25}$$

Consequently,

$$|\hat{J}_t - J_t^\star| \leq \lambda_1 \sqrt{\frac{\log(8(K-H)/\delta)}{2m}} + \lambda_2 \sqrt{3\sqrt{\frac{\log(8(K-H)/\delta)}{2m}}}. \tag{26}$$

Proposition 1 shows that the two ingredients used by STAWA are estimable uniformly over dense checkpoint sequences: validation loss for the plateau gate and stationarity for ranking inside that gate.

Proposition 2 (Finite-sample recovery of plateau membership)**.** *Under the assumptions of Proposition 1, define*

$$\eta_{n,\delta} \triangleq \sqrt{\frac{\log(4(K-H)/\delta)}{2n}}, \tag{27}$$

the population optimum

$$L_V^\star \triangleq \min_t R_V(p_t), \tag{28}$$

the population plateau

$$\mathcal{A}_\epsilon^\star \triangleq \{t : R_V(p_t) \leq L_V^\star + \epsilon\}, \tag{29}$$

and the empirical plateau

$$\widehat{\mathcal{A}}_\epsilon \triangleq \{t : \hat{L}_V(\theta_t) \leq \hat{L}_V^\star + \epsilon\}, \qquad \hat{L}_V^\star \triangleq \min_t \hat{L}_V(\theta_t). \tag{30}$$

Then with probability at least $1-\delta$, simultaneously for all t:

$$R_V(p_t) \leq L_V^\star + \epsilon - 2\eta_{n,\delta} \Longrightarrow t \in \widehat{\mathcal{A}}_\epsilon, \tag{31}$$

$$R_V(p_t) \geq L_V^\star + \epsilon + 2\eta_{n,\delta} \Longrightarrow t \notin \widehat{\mathcal{A}}_\epsilon. \tag{32}$$

Corollary 2 (Finite-sample recovery of within-plateau stationarity ranking)**.** *Under the assumptions of Proposition 1, define*

$$\eta_{J,m,\delta} \triangleq \lambda_1 \sqrt{\frac{\log(8(K-H)/\delta)}{2m}} + \lambda_2 \sqrt{3\sqrt{\frac{\log(8(K-H)/\delta)}{2m}}}. \tag{33}$$

If checkpoints s and t satisfy

$$s, t \in \mathcal{A}_{\epsilon - 2\eta_{n,\delta}}^\star \qquad \textit{and} \qquad J_s^\star + 2\eta_{J,m,\delta} < J_t^\star, \tag{34}$$

then with probability at least $1-\delta$,

$$s, t \in \widehat{\mathcal{A}}_\epsilon \qquad \textit{and} \qquad \hat{J}_s < \hat{J}_t. \tag{35}$$

Corollary 2 is the selector-consistency statement missing from purely philosophical stability arguments: once source-fit membership is recovered, the empirical post-hoc procedure ranks checkpoints correctly by stationarity inside the plateau whenever the population stationarity gap is large enough.

4.4 A target-risk certificate

We now combine the previous ingredients into the main bound.

Theorem 3 (Stationarity-sensitive target-risk certificate)**.** *Assume:*

(i) Assumption 1 holds.

(ii) The support set U and the labeled validation set V share the same source marginal P_S.

(iii) The target marginal P_T satisfies the density-ratio bound in Corollary 1.

(iv) For the local plateau family $\mathcal{F}_{t,H} \triangleq \{p_t, \bar{p}_{t,H}\}$,

$$\Gamma_t \triangleq \sup_{q \in \mathcal{F}_{t,H}} |R_T(q) - R_V(q)| \tag{36}$$

is finite.

Then for every checkpoint t,

$$R_T(p_t) \leq \hat{R}_V(p_t) + \Gamma_t + L_d \left(2\mu_t^S + \sigma_t^S \sqrt{\frac{1-\rho}{\rho}} \right) + \sqrt{\frac{\log(2(K-H)/\delta)}{2n}} \tag{37}$$

with probability at least $1-\delta$, simultaneously for all $t = 1, \ldots, K-H$.

Equation (37) is the main theoretical justification of STAWA. The uncontrollable term is still the source-target shift Γ_t. But among the terms the selector can influence, source-fit slack, mean drift, and drift spread are exactly the right objects. In particular, if $t \in \mathcal{A}_\epsilon^\star$, then $\hat{R}_V(p_t)$ is already constrained to lie near the best source-validation risk up to the plateau tolerance ϵ and estimation error, so the selector's remaining job is to minimize stationarity inside that plateau.

4.5 Why STAWA can see what SWAD cannot

The next theorem is the honest comparison result with flatness-only selectors.

Theorem 4 (Constructive separation from flatness-only selection)**.** *Fix $H = 1$ and use the absolute-value discrepancy $d(p,q) = |p-q|$ with squared loss $\ell(y,p) = (p-y)^2$. Let the input space be $\mathcal{X} = \{a, c\}$ with labels $y(a) = 0$ and $y(c) = 1$. Let:*

- *the labeled validation distribution V be the point mass on a,*
- *the support distribution U be uniform on $\{a, c\}$,*
- *the target distribution T be the point mass on c.*

Consider two checkpoints A and B with one-step future predictors A^+ and B^+ defined by

$$p_A(a) = 0, \quad p_A(c) = 1, \quad p_{A^+}(a) = 0, \quad p_{A^+}(c) = 1, \tag{38}$$

$$p_B(a) = 0, \quad p_B(c) = 0, \quad p_{B^+}(a) = 0, \quad p_{B^+}(c) = 1. \tag{39}$$

Assume further that the local labeled-validation loss landscapes around the two checkpoints are identical:

$$L_V^A(\theta + \Delta) = L_V^B(\theta + \Delta) = \frac{\beta}{2} \|\Delta\|_2^2 \qquad \text{for some } \beta > 0. \tag{40}$$

Then:

(i) $L_V(\theta_A) = L_V(\theta_B) = 0$ and any local flatness statistic derived from the labeled-validation loss is identical for A and B.

(ii) The STAWA *statistics are*

$$\mu_A = \sigma_A = 0, \qquad \mu_B = \frac{1}{2}, \qquad \sigma_B = \frac{1}{2}. \tag{41}$$

Hence for any $\lambda_1, \lambda_2 > 0$,

$$J_B - J_A = \frac{\lambda_1 + \lambda_2}{2} > 0. \tag{42}$$

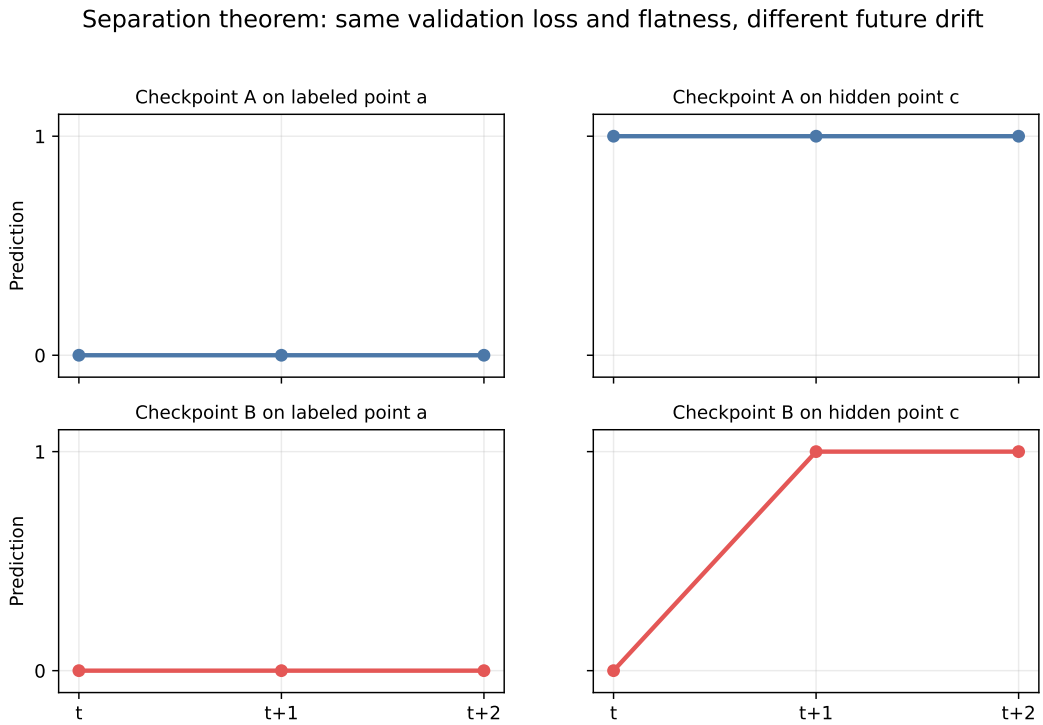

Figure 4: **Separation theorem intuition.** Two checkpoints can have the same labeled-validation loss and the same local flatness, yet one still contains a hidden support point whose prediction is being renegotiated by SGD. STAWA sees that hidden instability through future drift; a flatness-only selector does not.

(iii) The target risks satisfy

$$R_T(p_A) = 0, \qquad R_T(p_B) = 1. \tag{43}$$

Therefore any selector depending only on labeled-validation loss and local flatness is indifferent between A and B, while STAWA *strictly prefers the checkpoint with lower target risk inside any validation-loss plateau containing both.*

Theorem 4 does not say that STAWA universally beats SWAD. It says something narrower and more useful: there exist DG-relevant situations in which flatness-only selection is provably blind while a stationarity selector is not.

4.6 Dense checkpointing and low-drift averaging

Proposition 3 (Dense checkpoint selection converges to the continuous optimum)**.** *Let $\theta : [0, T] \to \mathbb{R}^p$ be a continuous training trajectory and let $s(t)$ be an L_s-Lipschitz score along the trajectory. Suppose checkpoints are saved at times $0 = t_1 < \cdots < t_K = T$ with maximum gap $\Delta = \max_k(t_{k+1} - t_k)$. Then*

$$\min_{1 \le k \le K} s(t_k) \le \min_{t \in [0,T]} s(t) + \frac{L_s \Delta}{2}. \tag{44}$$

Proposition 4 (Weight averaging is safe under local prediction linearity)**.** *Assume:*

(i) for every input x, the map $\theta \mapsto p_\theta(x)$ is twice differentiable on a convex neighborhood containing the selected window,

(ii) $\left\|\nabla^2_\theta p_\theta(x)\right\|_{\mathrm{op}} \le M_p$ on that neighborhood, and

(iii) $\ell(y, p)$ is convex in p and L_d-Lipschitz with respect to the prediction norm.

Let $\theta_W = \sum_{t \in W} w_t \theta_t$ with convex weights $\{w_t\}_{t \in W}$. Then for any distribution Q,

$$R_Q(\theta_W) \leq \sum_{t \in W} w_t R_Q(\theta_t) + \frac{L_d M_p}{2} \sum_{t \in W} w_t \left\| \theta_t - \theta_W \right\|_2^2 . \tag{45}$$

Proposition 4 is the local regularity statement needed to justify the final averaging step: if the selected window lies in a low-diameter locally linear region, weight averaging behaves like safe prediction averaging up to a controlled second-order remainder.

5 Experimental Protocol

This manuscript is complete as a method and theory paper draft, but all STAWA-specific experiments remain unrun. The sections below therefore specify the full evaluation protocol and the exact published numbers to beat.

5.1 Benchmarks and backbone

We will evaluate on the five standard DomainBed datasets used by SWAD and DiWA: PACS [11], VLCS [12], OfficeHome [13], TerraIncognita [14], and DomainNet [15]. Following prior work [1, 2, 16], the primary backbone is ResNet-50 and the primary protocol is leave-one-domain-out evaluation with model selection on held-out source validation data.

5.2 Baselines

We will compare against:

- ERM [16],
- CORAL [17],
- SWA [5],
- SWAD [1],
- DiWA [2],
- best validation checkpoint / early stopping,
- last checkpoint,
- EMA / moving-average checkpointing,
- lowest adjacent-drift checkpoint,
- latest plateau checkpoint,
- optional auxiliary comparisons to prediction-only checkpoint ensembles [4].

5.3 STAWA variants

The planned variants are:

- **STAWA-Mean**: $L_V + \lambda_1 \mu$,
- **STAWA-MeanVar**: $L_V + \lambda_1 \mu + \lambda_2 \sigma$ (default),
- **STAWA-Tail**: replace σ by an empirical top-quantile or CVaR drift term,
- **STAWA-Single**: best single checkpoint,
- **STAWA-Window**: contiguous plateau-constrained low-drift averaging (default).

We will also compare divergence choices:

- total variation / ℓ_1,
- Jensen–Shannon divergence,
- KL divergence,
- squared prediction difference for regression-style tasks.

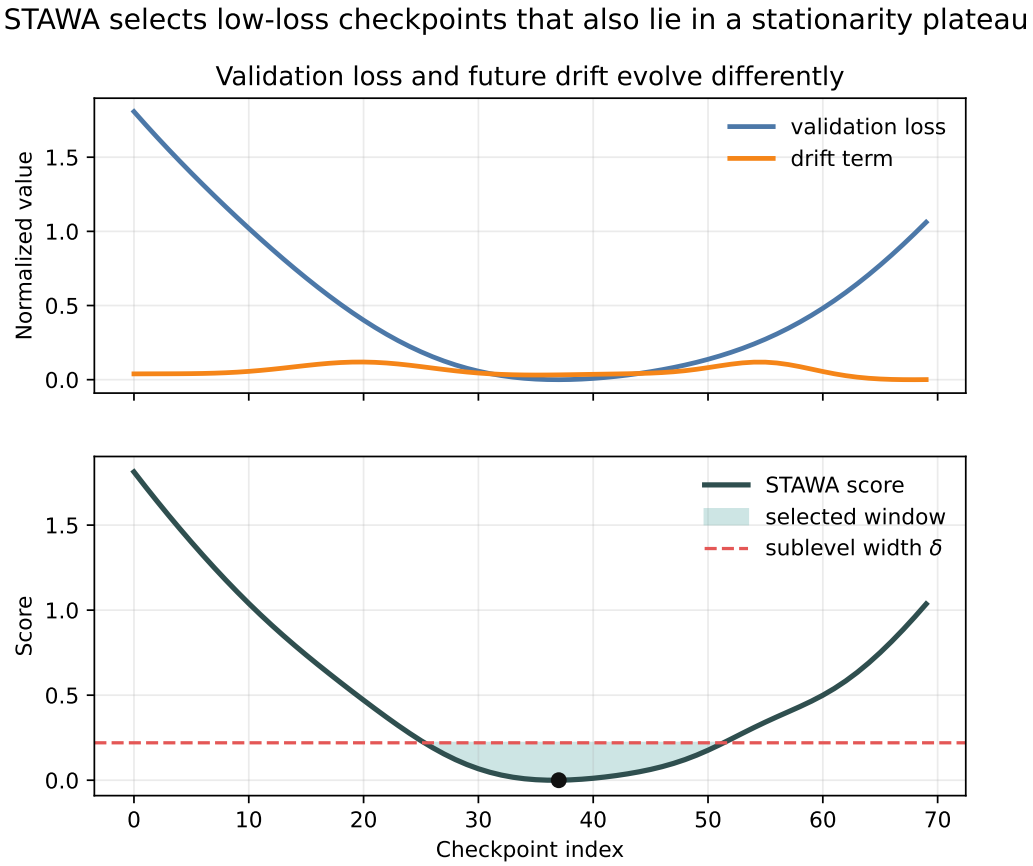

Figure 5: **Dense checkpointing and low-drift window selection.** STAWA preserves the same practical scaffold that made SWAD attractive: dense checkpointing and averaging within one connected plateau-constrained low-drift region. The difference is that the selector targets functional stationarity rather than parameter-space flatness.

5.4 Key falsifiable plots

Three figures are decisive for this paper:

1. checkpoint future drift versus final OOD accuracy correlation,
2. pairs of checkpoints with similar loss/flatness but different drift,
3. ablations showing that low drift is not just "later checkpoints," adjacent smoothness, or early stopping in disguise.

6 Published Reference Results and Placeholder Tables

We include exact published baselines from prior work so that the empirical target is explicit before running STAWA. Every new STAWA row remains intentionally marked as **TBD**.

7 Discussion and Limitations

What the theory proves. The theory proves six concrete points. First, mean predictive drift controls the risk gap between a checkpoint and its local prediction plateau. Second, the variance term controls hidden unstable subsets and bounded density-ratio shifts. Third, the empirical validation and stationarity components concentrate uniformly over dense checkpoint sequences. Fourth, finite

Table 1: **Published reference results on DomainBed with ResNet-50.** Numbers are copied from the cited papers and included here as reference targets for the planned STAWA evaluation.

Method	PACS	VLCS	OfficeHome	TerraInc	DomainNet	Avg
ERM [1, 2]	85.5	77.5	66.5	46.1	40.9	63.3
CORAL [2]	86.2	78.8	68.7	47.6	41.5	64.6
SWAD [1]	88.1	79.1	70.6	50.0	46.5	66.9
CORAL + SWAD [1]	88.3	78.9	71.3	51.0	46.8	67.3
DiWA (random init, $M = 60$) [2]	89.0	79.4	71.6	49.0	46.3	67.1
DiWA (LP init, $M = 60$) [2]	89.0	78.6	72.8	51.9	47.7	68.0
STAWA-Mean (ours)	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**
STAWA-MeanVar (ours)	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**
STAWA-Tail (ours)	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**

Table 2: **Published SWAD ablations copied from Cha et al. [1].** These results motivate dense checkpointing and contiguous window averaging, which STAWA preserves.

	Configuration				Out-of-domain			In-domain		
	t_s	t_e	lr	interval	PACS	VLCS	Avg.	PACS	VLCS	Avg.
$\text{SWA}_{\text{w/ cyclic}}$	4000	5000	Cyclic	100	85.9 $\pm$0.1	76.6 $\pm$0.1	81.2	97.1 $\pm$0.1	85.0 $\pm$0.2	91.0
$\text{SWA}_{\text{w/ const}}$	4000	5000	Const	100	86.5 $\pm$0.3	76.7 $\pm$0.2	81.6	97.3 $\pm$0.1	85.0 $\pm$0.2	91.1
$\text{SWAD}_{\text{w/o Dense}}$	Opt	Overfit	Const	100	86.5 $\pm$0.4	78.0 $\pm$0.7	82.2	97.6 $\pm$0.1	85.8 $\pm$0.4	91.7
$\text{SWAD}_{\text{w/o Opt-Overfit}}$	4000	5000	Const	1	86.6 $\pm$0.6	76.9 $\pm$0.3	81.7	97.5 $\pm$0.1	85.2 $\pm$0.1	91.3
$\text{SWAD}_{\text{w/o Overfit}}$	Opt	5000	Const	1	87.1 $\pm$0.3	77.6 $\pm$0.1	82.4	97.7 $\pm$0.1	85.8 $\pm$0.3	91.8
$\text{SWAD}_{\text{fit-on-val}}$	Val	Val	Const	1	86.2 $\pm$0.2	78.6 $\pm$0.1	82.4	97.5 $\pm$0.2	85.8 $\pm$0.3	91.7
SWAD (proposed)	Opt	Overfit	Const	1	87.1 $\pm$0.2	78.9 $\pm$0.2	83.0	97.7 $\pm$0.1	86.1 $\pm$0.5	91.9

samples recover plateau membership and within-plateau stationarity ranking whenever the population margins are large enough. Fifth, the target-risk certificate depends on source-fit slack, mean drift, and drift spread as the controllable terms. Sixth, flatness-only selectors can be provably blind on an explicit class of examples.

What the theory does not prove. The theory does not prove universal empirical superiority over SWAD, DiWA, or any other DG method. It cannot. It also does not prove that every low-drift checkpoint is semantically robust. The value of the theory is narrower and more honest: it identifies a failure mode of flatness-only selection, gives a measurable function-space alternative, and shows why that alternative is a principled selector inside one completed trajectory.

The main practical risk. Low future drift is not automatically the same as good OOD generalization. A predictor can be stably wrong. The paper's empirical burden is therefore clear: it must show that functionally stationary checkpoints correlate with OOD quality more strongly than flatness-only signals do.

The main competing explanation. The most serious hostile-review interpretation is that STAWA merely rediscovers late-phase checkpoint selection. That is exactly why the baseline suite must include best validation, last checkpoint, EMA, adjacent-drift selection, and latest-plateau heuristics. If those simpler timing rules explain the same gains, then STAWA is not yet a new DG signal.

Why the future-looking stationarity score is still acceptable. STAWA is explicitly a post-hoc trajectory selector. Its stationarity score depends on future checkpoints inside one completed run. That is not a bug in this setting; it is the defining object. The method would only become problematic if it were sold as an online training rule, which this paper does not do.

Current limitation of this manuscript. The obvious limitation is empirical: the STAWA experiments have not yet been run. The manuscript is therefore complete as a theory-and-method paper draft, but not complete as a submission-ready empirical claim until the **TBD** sections are filled.

Table 3: **Planned main results table.** Every STAWA cell remains intentionally left as **TBD** pending experiments.

Method	PACS	VLCS	OfficeHome	TerraInc	DomainNet	Avg
SWAD	88.1	79.1	70.6	50.0	46.5	66.9
DiWA (best published)	89.0	78.6	72.8	51.9	47.7	68.0
STAWA-Mean	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**
STAWA-MeanVar	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**
STAWA-Tail	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**

Table 4: **Planned STAWA ablations.** These rows are intentionally blank pending experiments.

Ablation	Avg
Mean only vs. mean + variance	**TBD**
Mean + variance vs. CVaR tail term	**TBD**
TV vs. JS vs. KL drift	**TBD**
Single checkpoint vs. plateau-constrained low-drift window	**TBD**
Horizon length H	**TBD**
Best-val / last / EMA / adjacent-drift / latest-plateau baselines	**TBD**
Drift/OOD correlation	**TBD**
Compute overhead vs. SWAD	**TBD**

8 Conclusion

We proposed STAWA, a label-free single-run retrospective DG selector based on functional stationarity rather than pooled flatness. The method first restricts to a near-best source-loss plateau, then ranks that plateau by mean and spread of short-horizon predictive drift, and finally averages a contiguous plateau-constrained low-drift window. Our analysis shows that low drift places a checkpoint near its local plateau, that variance controls hidden unstable subsets, that the empirical components concentrate and recover plateau membership plus within-plateau ranking, and that flatness-only selectors can miss target-relevant instability that STAWA captures. The empirical results remain to be run, but the manuscript now contains the method, the theory, the figures, the protocol, and the published baselines needed to test whether functional stationarity is a genuinely new selector principle rather than a repackaging of late-phase checkpoint smoothing.

References

[1] Junbum Cha, Sanghyuk Chun, Kyungjae Lee, Han-Cheol Cho, Seunghyun Park, Yunsung Lee, and Sungrae Park. SWAD: Domain generalization by seeking flat minima. In *Advances in Neural Information Processing Systems*, 2021.

[2] Alexandre Ramé, Matthieu Kirchmeyer, Thibaud Rahier, Alain Rakotomamonjy, Patrick Gallinari, and Matthieu Cord. Diverse weight averaging for out-of-distribution generalization. In *Advances in Neural Information Processing Systems*, 2022.

[3] Nisha Chandramoorthy, Andreas Loukas, Khashayar Gatmiry, and Stefanie Jegelka. On the generalization of learning algorithms that do not converge. *arXiv preprint arXiv:2208.07951*, 2022.

[4] Hugh Chen, Scott Lundberg, and Su-In Lee. Checkpoint ensembles: Ensemble methods from a single training process. *arXiv preprint arXiv:1710.03282*, 2017.

[5] Pavel Izmailov, Dmitrii Podoprikhin, Timur Garipov, Dmitry Vetrov, and Andrew Gordon Wilson. Averaging weights leads to wider optima and better generalization. In *Conference on Uncertainty in Artificial Intelligence*, 2018.

[6] Timur Garipov, Pavel Izmailov, Dmitrii Podoprikhin, Dmitry Vetrov, and Andrew Gordon Wilson. Loss surfaces, mode connectivity, and fast ensembling of DNNs. In *Advances in Neural Information Processing Systems*, 2018.

[7] Jonathan Frankle, Gintare Karolina Dziugaite, Daniel M. Roy, and Michael Carbin. Linear mode connectivity and the lottery ticket hypothesis. In *International Conference on Machine Learning*, 2020.

[8] Mitchell Wortsman, Gabriel Ilharco, Samir Yitzhak Gadre, Rebecca Roelofs, Raphael Gontijo-Lopes, Ari S. Morcos, Hongseok Namkoong, Ali Farhadi, Yair Carmon, Simon Kornblith, and Ludwig Schmidt. Model soups: Averaging weights of multiple fine-tuned models improves accuracy without increasing inference time. In *International Conference on Machine Learning*, 2022.

[9] Olivier Bousquet and André Elisseeff. Stability and generalization. *Journal of Machine Learning Research*, 2:499–526, 2002.

[10] Moritz Hardt, Benjamin Recht, and Yoram Singer. Train faster, generalize better: Stability of stochastic gradient descent. In *International Conference on Machine Learning*, 2016.

[11] Da Li, Yongxin Yang, Yi-Zhe Song, and Timothy Hospedales. Deeper, broader and artier domain generalization. In *IEEE International Conference on Computer Vision*, 2017.

[12] Chen Fang, Ye Xu, and Daniel Rockmore. Unbiased metric learning: On the utilization of multiple datasets and web images for softening bias. In *IEEE International Conference on Computer Vision*, 2013.

[13] Hemanth Venkateswara, Jose Eusebio, Shayok Chakraborty, and Sethuraman Panchanathan. Deep hashing network for unsupervised domain adaptation. In *IEEE Conference on Computer Vision and Pattern Recognition*, 2017.

[14] Sara Beery, Grant Van Horn, and Pietro Perona. Recognition in terra incognita. In *European Conference on Computer Vision*, 2018.

[15] Xingchao Peng, Qinxun Bai, Xide Xia, Zijun Huang, Kate Saenko, and Bo Wang. Moment matching for multi-source domain adaptation. In *IEEE International Conference on Computer Vision*, 2019.

[16] Ishaan Gulrajani and David Lopez-Paz. In search of lost domain generalization. In *International Conference on Learning Representations*, 2021.

[17] Baochen Sun and Kate Saenko. Deep CORAL: Correlation alignment for deep domain adaptation. In *European Conference on Computer Vision Workshops*, 2016.

A Detailed Proofs

A.1 Proof of Theorem 1

Proof. We first prove Eq. (15). Fix an input x. By definition,

$$\bar{p}_{t,H}(x) - p_t(x) = \frac{1}{H+1} \sum_{h=1}^{H} \left(p_{t+h}(x) - p_t(x) \right). \tag{46}$$

Since d is induced by a norm, the triangle inequality gives

$$d(p_t(x), \bar{p}_{t,H}(x)) = \left\| \frac{1}{H+1} \sum_{h=1}^{H} \left(p_{t+h}(x) - p_t(x) \right) \right\| \tag{47}$$

$$\leq \frac{1}{H+1} \sum_{h=1}^{H} d(p_t(x), p_{t+h}(x)) \tag{48}$$

$$\leq \frac{1}{H} \sum_{h=1}^{H} d(p_t(x), p_{t+h}(x)) = \Delta_t(x). \tag{49}$$

By Assumption 1,

$$\ell(y, p_t(x)) \leq \ell(y, \bar{p}_{t,H}(x)) + L_d \Delta_t(x). \tag{50}$$

Taking expectation over $(x, y) \sim Q$ yields

$$R_Q(p_t) \leq R_Q(\bar{p}_{t,H}) + L_d \mu_t^Q. \tag{51}$$

This proves Eq. (15).

For Eq. (17), fix (x, y). By convexity of $\ell(y, \cdot)$,

$$\ell(y, \bar{p}_{t,H}(x)) = \ell\left(y, \frac{1}{H+1} \sum_{h=0}^{H} p_{t+h}(x)\right) \leq \frac{1}{H+1} \sum_{h=0}^{H} \ell(y, p_{t+h}(x)). \tag{52}$$

Taking expectation over $(x, y) \sim Q$ proves

$$R_Q(\bar{p}_{t,H}) \leq \frac{1}{H+1} \sum_{h=0}^{H} R_Q(p_{t+h}). \tag{53}$$

□

A.2 Proof of Theorem 2

Proof. Let I_A denote the indicator of the subset A. Since $P(A) = \rho$,

$$\mathbb{E}[\Delta \mid A] = \frac{\mathbb{E}[\Delta I_A]}{\rho}. \tag{54}$$

Now decompose

$$\mathbb{E}[\Delta I_A] = \mathbb{E}[\Delta]\mathbb{E}[I_A] + \mathrm{Cov}(\Delta, I_A) \tag{55}$$

$$= \rho\mu + \mathrm{Cov}(\Delta, I_A). \tag{56}$$

By Cauchy–Schwarz,

$$\mathrm{Cov}(\Delta, I_A) \leq \sqrt{\mathrm{Var}(\Delta)\,\mathrm{Var}(I_A)} = \sigma\sqrt{\rho(1-\rho)}. \tag{57}$$

Therefore

$$\mathbb{E}[\Delta I_A] \leq \rho\mu + \sigma\sqrt{\rho(1-\rho)}. \tag{58}$$

Dividing by ρ gives

$$\mathbb{E}[\Delta \mid A] \leq \mu + \sigma\sqrt{\frac{1-\rho}{\rho}}. \tag{59}$$

□

A.3 Proof of Corollary 1

Proof. Let

$$w(x) \triangleq \frac{dP_T}{dP_U}(x). \tag{60}$$

Then $w \geq 0$, $\mathbb{E}_{P_U}[w] = 1$, and $w \leq 1/\rho$ almost surely. The target drift can be written as

$$\mu_t^T = \mathbb{E}_{P_T}[\Delta_t(x)] = \mathbb{E}_{P_U}[w(x)\Delta_t(x)]. \tag{61}$$

Decompose

$$\mathbb{E}_{P_U}[w\Delta_t] = \mathbb{E}_{P_U}[w]\mathbb{E}_{P_U}[\Delta_t] + \mathrm{Cov}_{P_U}(w, \Delta_t) \tag{62}$$

$$= \mu_t^U + \mathrm{Cov}_{P_U}(w, \Delta_t). \tag{63}$$

Again by Cauchy–Schwarz,

$$\mathrm{Cov}_{P_U}(w, \Delta_t) \leq \sqrt{\mathrm{Var}_{P_U}(w)}\,\sigma_t^U. \tag{64}$$

It remains to bound the variance of w. Since $0 \le w \le 1/\rho$ and $\mathbb{E}[w] = 1$,

$$\mathbb{E}[w^2] \le \frac{1}{\rho}\mathbb{E}[w] = \frac{1}{\rho}. \tag{65}$$

Therefore

$$\mathrm{Var}(w) = \mathbb{E}[w^2] - \mathbb{E}[w]^2 \le \frac{1}{\rho} - 1 = \frac{1-\rho}{\rho}. \tag{66}$$

Combining the last two displays yields

$$\mu_t^T \le \mu_t^U + \sigma_t^U \sqrt{\frac{1-\rho}{\rho}}. \tag{67}$$

□

A.4 Proof of Proposition 1

Proof. The validation loss is bounded in $[0, 1]$, so Hoeffding's inequality gives, for each t,

$$\Pr\Big(\Big|\hat{L}_V(\theta_t) - R_V(p_t)\Big| > \epsilon\Big) \le 2e^{-2n\epsilon^2}. \tag{68}$$

Applying a union bound over $t = 1, \ldots, K - H$ proves Eq. (23).

Next, each drift variable $\Delta_t(x_i)$ lies in $[0, 1]$, so another application of Hoeffding plus a union bound gives Eq. (24).

For the standard deviation term, define

$$m_{2,t} \triangleq \mathbb{E}[(\Delta_t)^2], \qquad \hat{m}_{2,t} \triangleq \frac{1}{m}\sum_{i=1}^{m} \Delta_t(x_i)^2. \tag{69}$$

Since $\Delta_t(x_i)^2 \in [0, 1]$, Hoeffding and a union bound yield

$$|\hat{m}_{2,t} - m_{2,t}| \le \sqrt{\frac{\log(8(K-H)/\delta)}{2m}} \tag{70}$$

simultaneously for all t with probability at least $1 - \delta/2$.

Now

$$\sigma_t^2 = m_{2,t} - (\mu_t^U)^2, \qquad \hat{\sigma}_t^2 = \hat{m}_{2,t} - (\hat{\mu}_t)^2. \tag{71}$$

Hence

$$\begin{aligned} |\hat{\sigma}_t^2 - \sigma_t^2| &\le |\hat{m}_{2,t} - m_{2,t}| + |(\hat{\mu}_t)^2 - (\mu_t^U)^2| && (72)\\ &\le |\hat{m}_{2,t} - m_{2,t}| + |\hat{\mu}_t - \mu_t^U|\,|\hat{\mu}_t + \mu_t^U| && (73)\\ &\le |\hat{m}_{2,t} - m_{2,t}| + 2|\hat{\mu}_t - \mu_t^U|, && (74) \end{aligned}$$

because both means lie in $[0, 1]$. Using the previous bounds gives

$$|\hat{\sigma}_t^2 - \sigma_t^2| \le 3\sqrt{\frac{\log(8(K-H)/\delta)}{2m}}. \tag{75}$$

Finally, since $u \mapsto \sqrt{u}$ is $1/2$-Hölder on $[0, \infty)$,

$$|\hat{\sigma}_t - \sigma_t| \le \sqrt{|\hat{\sigma}_t^2 - \sigma_t^2|} \le \sqrt{3\sqrt{\frac{\log(8(K-H)/\delta)}{2m}}}. \tag{76}$$

Combining the bounds on $\hat{\mu}_t$ and $\hat{\sigma}_t$ proves Eq. (26); Eq. (23) is already the validation-loss component bound needed for the plateau gate. □

A.5 Proof of Proposition 2

Proof. By Proposition 1, with probability at least $1-\delta$ we have simultaneously

$$|\hat{L}_V(\theta_u) - R_V(p_u)| \le \eta_{n,\delta} \qquad \text{for every } u \in \{1,\dots,K-H\}. \tag{77}$$

Fix that event. Let $u^\star$ be a minimizer of $R_V(p_u)$, so $R_V(p_{u^\star}) = L_V^\star$. Then

$$\hat{L}_V^\star \le \hat{L}_V(\theta_{u^\star}) \le L_V^\star + \eta_{n,\delta}. \tag{78}$$

Now suppose

$$R_V(p_t) \le L_V^\star + \epsilon - 2\eta_{n,\delta}. \tag{79}$$

Then

$$\hat{L}_V(\theta_t) \le R_V(p_t) + \eta_{n,\delta} \le L_V^\star + \epsilon - \eta_{n,\delta} \le \hat{L}_V^\star + \epsilon, \tag{80}$$

so $t \in \widehat{\mathcal{A}}_\epsilon$.

Conversely, if

$$R_V(p_t) \ge L_V^\star + \epsilon + 2\eta_{n,\delta}, \tag{81}$$

then

$$\hat{L}_V(\theta_t) \ge R_V(p_t) - \eta_{n,\delta} \ge L_V^\star + \epsilon + \eta_{n,\delta} \ge \hat{L}_V^\star + \epsilon, \tag{82}$$

so $t \notin \widehat{\mathcal{A}}_\epsilon$. This proves both implications. □

A.6 Proof of Corollary 2

Proof. By Proposition 1, with probability at least $1-\delta$ we have simultaneously

$$|\hat{J}_u - J_u^\star| \le \eta_{J,m,\delta} \qquad \text{for every } u \in \{1,\dots,K-H\}. \tag{83}$$

On the same event, Proposition 2 implies that every checkpoint in $\mathcal{A}^\star_{\epsilon-2\eta_{n,\delta}}$ belongs to $\widehat{\mathcal{A}}_\epsilon$. Hence $s,t \in \widehat{\mathcal{A}}_\epsilon$.

Finally,

$$\hat{J}_s \le J_s^\star + \eta_{J,m,\delta} < J_t^\star - \eta_{J,m,\delta} \le \hat{J}_t, \tag{84}$$

where the strict middle inequality uses the assumed population margin

$$J_s^\star + 2\eta_{J,m,\delta} < J_t^\star. \tag{85}$$

Therefore $\hat{J}_s < \hat{J}_t$ with probability at least $1-\delta$. □

A.7 Proof of Theorem 3

Proof. By Theorem 1,

$$R_T(p_t) \le R_T(\bar{p}_{t,H}) + L_d \mu_t^T. \tag{86}$$

By definition of Γ_t,

$$R_T(\bar{p}_{t,H}) \le R_V(\bar{p}_{t,H}) + \Gamma_t. \tag{87}$$

Applying Theorem 1 again, now on the source distribution V,

$$R_V(\bar{p}_{t,H}) \le R_V(p_t) + L_d \mu_t^S. \tag{88}$$

Combining Eqs. (86)–(88) gives

$$R_T(p_t) \le R_V(p_t) + \Gamma_t + L_d(\mu_t^S + \mu_t^T). \tag{89}$$

By Corollary 1,

$$\mu_t^T \le \mu_t^S + \sigma_t^S \sqrt{\frac{1-\rho}{\rho}}. \tag{90}$$

Substituting this into Eq. (89) yields

$$R_T(p_t) \le R_V(p_t) + \Gamma_t + L_d\left(2\mu_t^S + \sigma_t^S\sqrt{\frac{1-\rho}{\rho}}\right). \tag{91}$$

Finally, because the validation loss is bounded in $[0,1]$, Hoeffding's inequality with a union bound over $t = 1,\dots,K-H$ gives

$$R_V(p_t) \le \hat{R}_V(p_t) + \sqrt{\frac{\log(2(K-H)/\delta)}{2n}} \tag{92}$$

simultaneously for all t with probability at least $1-\delta$. Combining the last two displays proves Eq. (37). □

A.8 Proof of Theorem 4

Proof. Part (i) is immediate. The labeled validation set contains only the point a with label 0, and both checkpoints predict 0 on that point. Therefore

$$L_V(\theta_A) = L_V(\theta_B) = 0. \tag{93}$$

By construction, the local labeled-validation loss landscapes are identical quadratic functions, so any local flatness statistic computed from them is also identical.

For part (ii), checkpoint A is perfectly stationary over the one-step horizon on both support points:

$$\Delta_A(a) = |p_A(a) - p_{A^+}(a)| = 0, \qquad \Delta_A(c) = |p_A(c) - p_{A^+}(c)| = 0. \tag{94}$$

Hence $\mu_A = 0$ and $\sigma_A = 0$.

Checkpoint B is stationary on a but not on c:

$$\Delta_B(a) = |0 - 0| = 0, \qquad \Delta_B(c) = |0 - 1| = 1. \tag{95}$$

Since U is uniform on $\{a, c\}$,

$$\mu_B = \frac{0+1}{2} = \frac{1}{2}. \tag{96}$$

The variance is

$$\sigma_B^2 = \frac{1}{2}\left(0 - \frac{1}{2}\right)^2 + \frac{1}{2}\left(1 - \frac{1}{2}\right)^2 = \frac{1}{4}, \tag{97}$$

so $\sigma_B = \frac{1}{2}$. Therefore

$$S_B - S_A = \lambda_1\left(\frac{1}{2}\right) + \lambda_2\left(\frac{1}{2}\right) = \frac{\lambda_1 + \lambda_2}{2} > 0. \tag{98}$$

For part (iii), the target distribution is concentrated on c with label 1. Under squared loss,

$$R_T(p_A) = (1-1)^2 = 0, \qquad R_T(p_B) = (0-1)^2 = 1. \tag{99}$$

This proves the strict target-risk separation. □

A.9 Proof of Proposition 3

Proof. Let $t^\star \in \arg\min_{t\in[0,T]} s(t)$. By construction of the grid, there exists some checkpoint time t_k satisfying $|t_k - t^\star| \le \Delta/2$. Since s is L_s-Lipschitz,

$$s(t_k) \le s(t^\star) + L_s|t_k - t^\star| \le s(t^\star) + \frac{L_s\Delta}{2}. \tag{100}$$

Taking the minimum over k proves the claim. □

A.10 Proof of Proposition 4

Proof. Let

$$\bar{p}_W(x) \triangleq \sum_{t\in W} w_t p_{\theta_t}(x) \tag{101}$$

be the prediction-space average over the selected window, and let

$$\theta_W \triangleq \sum_{t\in W} w_t \theta_t. \tag{102}$$

Fix an input x. By Taylor's theorem around θ_W, for each $t \in W$ there exists a point $\xi_t(x)$ on the segment between θ_W and θ_t such that

$$p_{\theta_t}(x) = p_{\theta_W}(x) + \nabla_\theta p_{\theta_W}(x)^\top(\theta_t - \theta_W) + \frac{1}{2}(\theta_t - \theta_W)^\top \nabla_\theta^2 p_{\xi_t(x)}(x)(\theta_t - \theta_W). \tag{103}$$

Multiply by w_t and sum over t. The linear terms cancel because

$$\sum_{t\in W} w_t(\theta_t - \theta_W) = 0. \tag{104}$$

Hence

$$\bar{p}_W(x) - p_{\theta_W}(x) = \frac{1}{2} \sum_{t \in W} w_t (\theta_t - \theta_W)^\top \nabla^2_\theta p_{\xi_t(x)}(x)(\theta_t - \theta_W). \tag{105}$$

Taking norms and using the Hessian bound gives

$$d(\bar{p}_W(x), p_{\theta_W}(x)) \leq \frac{M_p}{2} \sum_{t \in W} w_t \left\|\theta_t - \theta_W\right\|_2^2. \tag{106}$$

By convexity of $\ell(y, \cdot)$,

$$\ell(y, \bar{p}_W(x)) \leq \sum_{t \in W} w_t \ell(y, p_{\theta_t}(x)). \tag{107}$$

By L_d-Lipschitz continuity,

$$\ell(y, p_{\theta_W}(x)) \leq \ell(y, \bar{p}_W(x)) + L_d \, d(\bar{p}_W(x), p_{\theta_W}(x)). \tag{108}$$

Substituting Eq. (106) yields

$$\ell(y, p_{\theta_W}(x)) \leq \sum_{t \in W} w_t \ell(y, p_{\theta_t}(x)) + \frac{L_d M_p}{2} \sum_{t \in W} w_t \left\|\theta_t - \theta_W\right\|_2^2. \tag{109}$$

Taking expectation over $(x, y) \sim Q$ proves Eq. (45). □

B Implementation Notes

Choice of drift metric. The cleanest theory uses a norm-induced discrepancy such as ℓ_1 or total variation. In practice, Jensen–Shannon or KL drift may correlate better with accuracy for classification. The theory then applies after translating those divergences to total variation through standard inequalities or after clipping probabilities.

Horizon length. The horizon H should be long enough to expose unresolved function drift but short enough to stay inside one local training phase. The main paper will treat H as a small hyperparameter and report sensitivity.

Why a contiguous window? Contiguity is inherited from SWA and SWAD. It is not merely aesthetic: it keeps the selected checkpoints inside one local region of the trajectory, which is exactly the regime in which Proposition 4 is most credible.